Mudit Todi

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Education

LNMIIT Jaipur, Rajasthan
B.Tech in Electronics & Communication Engineering — CGPA: 7.4 *Jul. 2023 – Jun. 2027*

Experience

PCB Design Intern Feb. 2026
S7 Innovation Labs LLP *Remote*

- Designed **2 production PCBs** for SEVEN robot, covering battery charging, power distribution, MCU control, sensing, audio, display, and storage interfaces.
- Designed the charging PCB with power-path management, reverse-polarity protection, TVS surge clamping, and filtered system power using **EasyEDA**.
- Integrated **5+ peripheral interfaces** on the main-controller PCB, including I^2S audio, I^2C sensing, and shared SPI peripherals; applied DFM and ESD protection practices.

Projects

4-DOF Robotic Arm — Robust Direct MRAC GitHub | Apr. 2026
ROS 2 Jazzy, C++, MoveIt 2, Gazebo Harmonic

- Built a **4-DOF Robust MRAC** controller in C++ with online feedback, feedforward, and disturbance adaptation.
- Implemented Lyapunov-based adaptation with σ-modification at a **1 kHz** closed-loop control rate.
- Integrated **ROS 2, ros2_control, MoveIt 2, and Gazebo** with OMPL/KDL planning and Hermite trajectory interpolation.

Nonlinear & Adaptive Control Benchmark — Cart-Pole GitHub | Mar. 2026
Python, MuJoCo, Gymnasium, NumPy, Matplotlib

- Derived nonlinear cart-pole dynamics using **Lagrangian mechanics** and designed a cascaded position–angle controller.
- Analytically tuned **IMC-PID** and implemented Direct MRAC / Robust MRAC with online gain adaptation.
- Achieved **43.7% lower ISE** with Robust MRAC under disturbance versus the baseline controller, while benchmarking recovery time and control effort.

Drone-to-Ground LiFi / Laser Communication System GitHub | IMC 2025
Pixhawk, ArduPilot, Embedded Systems, Optical Communication

- Designed IR Tx/Rx modules achieving **830 Kbps at 6 ft** and **52 Kbps at 20 ft** over a free-space optical link.
- Deployed a YOLOv8 weapon detection model on a Raspberry Pi 5 for edge-based security surveillance.
- Demonstrated the complete drone-to-ground communication system at the **IMC 2025** technology exhibition.

Technical Skills

Robotics: ROS 2 Jazzy, MoveIt 2, Gazebo Harmonic, ros2_control, URDF/Xacro, RViz, OMPL, KDL

Control: MRAC, Robust MRAC, Adaptive Control, IMC-PID, Lyapunov Analysis, Disturbance Rejection

Programming: C++, Python, MATLAB, NumPy, Matplotlib

Simulation: MuJoCo, Gymnasium, Nonlinear Modeling, Trajectory Generation

Embedded & UAV: Pixhawk, ArduPilot, QGroundControl, ESP32, UART/I^2C/SPI

Hardware: PCB Design, KiCad, EasyEDA, PCB Layout, DFM

Achievements

ISRO START Program: Certificate of Merit — Grade A (2025) **e-Yantra Krishi Drone:** Stage 2 Qualifier

CBSE Mathematics: Certificate of Merit (2020)